Lab10

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2. Intro to RCSB PDB

PDB Statistics:

```
stats <- read.csv("pdb_stats.csv")
```

> **Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.**

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
stats.summarized <- stats %>%
  summarize(X.ray = sum(X.ray), EM = sum(EM), Total = sum(Total))

Ans1 <- (stats.summarized$X.ray +
           stats.summarized$EM)/stats.summarized$Total * 100
Ans1
```

```
[1] 93.7892
```

93.7892% are solved by X-ray or EM.

Q2: What proportion of structures in the PDB are protein?

```r
Ans2 <- sum(c(stats[1, "Total"], stats[2, "Total"], stats[3, "Total"])
            )/stats.summarized$Total * 100
Ans2
```

```
[1] 97.9118
```

97.9118% of structures in the PDB are proteins, including protein only, protein-oligosaccharide, and protein-nucleic acid.

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There were 1,227 Structures when "HIV-1 Protease" was used as the query.

3. Visualizing the HIV-1 protease structure

Using Mol*

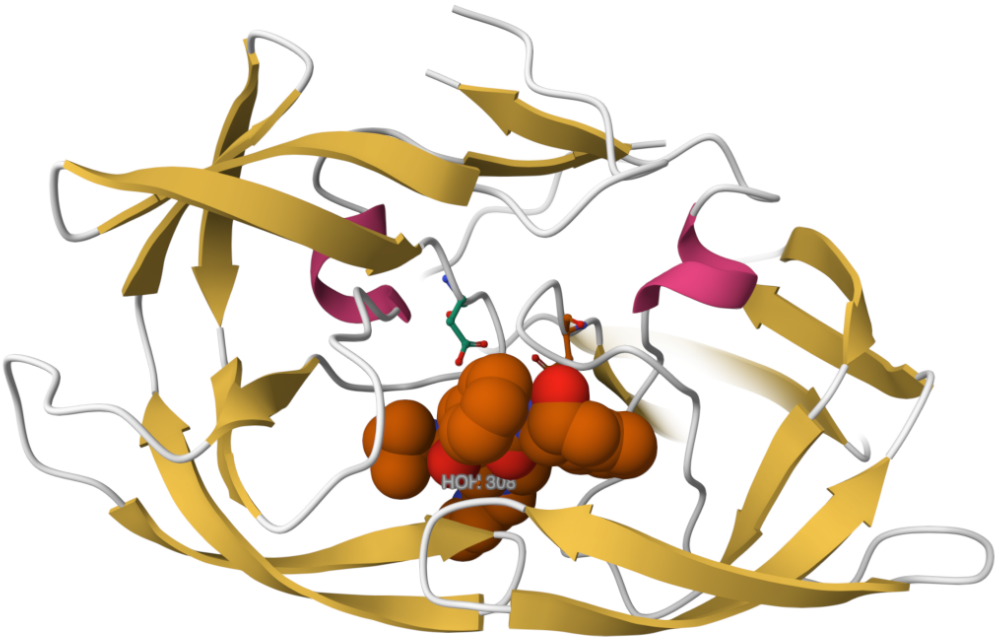


Figure 1: Mol* Visualization of 1HSG

Delving Deeper

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Hydrogens are probably too small to resolve using X-ray crystallography on their own, and in all these structures they appear bound to another, larger atom - usually oxygen or nitrogen. In the water molecules, we only see one atom for the same reason. This actually represents an oxygen bound to two hydrogens, but the hydrogens are too small to resolve from the mugh larger (in comparison to the hydrogen) central oxygen.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Yes, the water molecule was HOH 308 and coordinates with the ligand.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

Figure 2: Mol* Visualisation of 1HSG with ball and stick representations of D25 on both chains, and a label on the critical co-ordinating water molecule.

[Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

The antiparallel beta-sheet between the C-terminals of both chains, specifically residues T96, L97, and N98, as seen in the figure below.

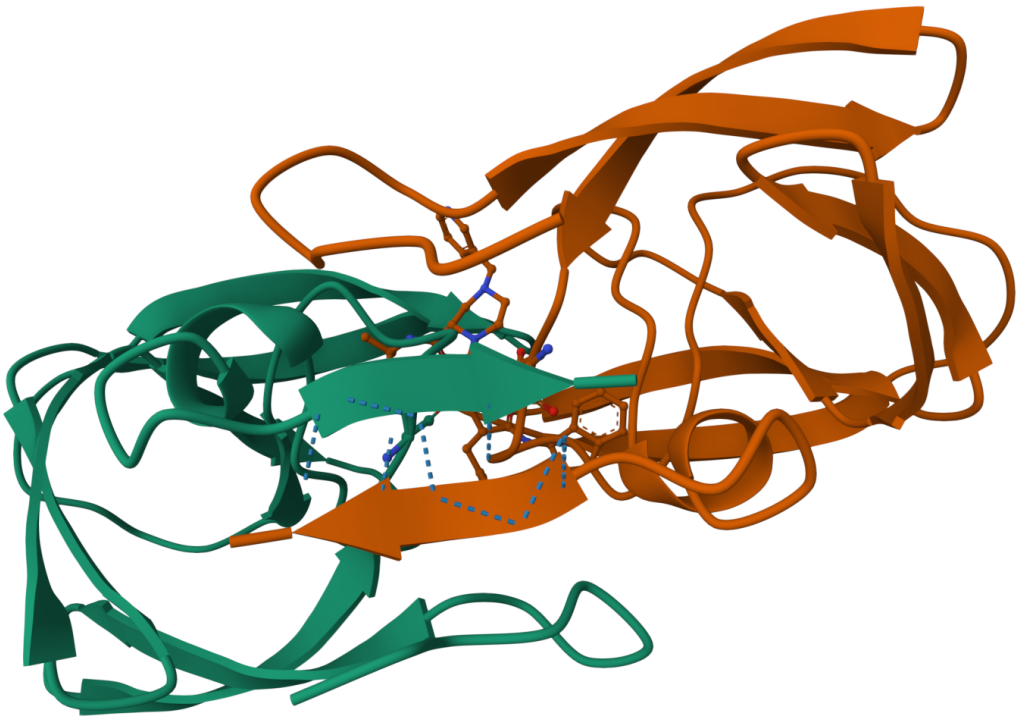


Figure 3: Antiparallel beta sheet(shown in green and orange arrows) between C-terminal of chain A and C-terminal of chain B of HIV-1 Protease that can only form in the dimer.

4. Intro to Bio3D in R

```
library(bio3d)
```

Reading in the PDB file data

```
pdb <- read.pdb("1hsg")
```

```
  Note:  Accessing on-line PDB file
```

```r
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

2

```r
attributes(pdb)
```

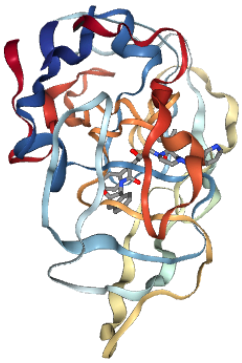
```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```r
head(pdb$atom)
```

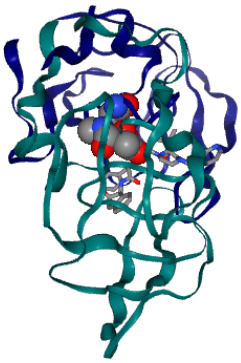
```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

Quick PDB visualization in R

```r
library(bio3dview)
library(NGLVieweR)

view.pdb(pdb) %>% setSpin()
```



```r
select <- atom.select(pdb, resno = 25)

view.pdb(pdb, cols = c("navy", "teal"),
         highlight = select,
         highlight.style = "spacefill") %>%
  setRock()
```



Predicting functional motions of a single structure

```
adk <- read.pdb("6S36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
adk
```

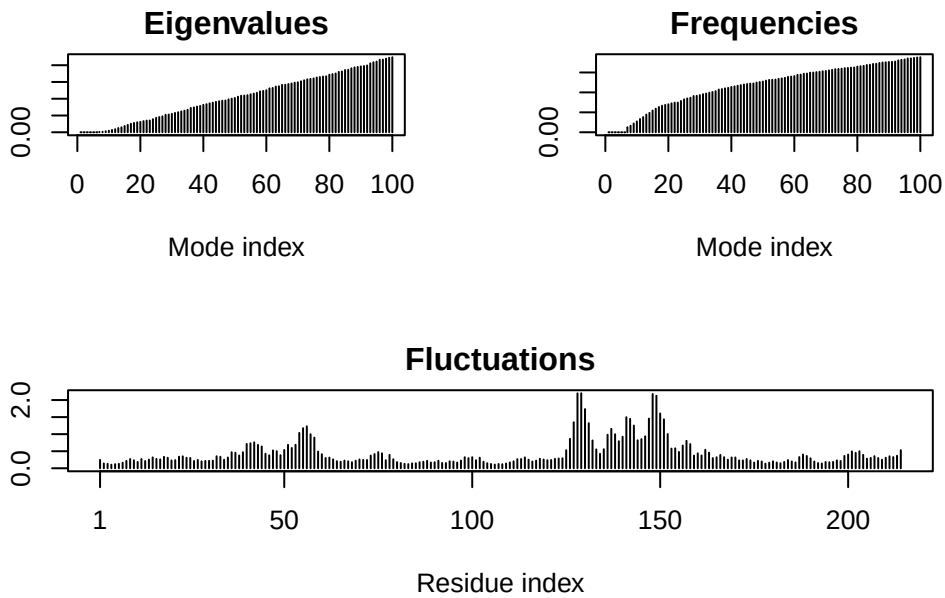
```
 Call:  read.pdb(file = "6S36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

Performing flexibility prediction - Normal Mode Analysis (NMA)

```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.011 seconds.
 Diagonalizing Hessian...   Done in 0.529 seconds.
```

```r
plot(m)
```



Making a "movie" of the predictions using molecular trajectiory - mktrj()

```r
mktrj(m, file = "adk_m7.pdb")
```

```r
view.nma(m, pdb = adk)
```

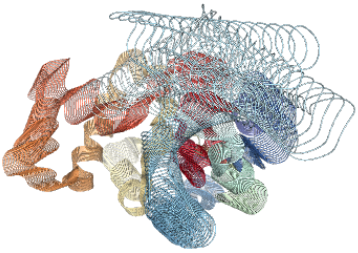


Figure 4: Output of visualizing the adk_m7 file in PDB

5. Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

TRUE

Search and Retrieve ADK Structures

```
aa <- get.seq("1ake_a")
```

```
Warning in get.seq("1ake_a"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.
```

```
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214

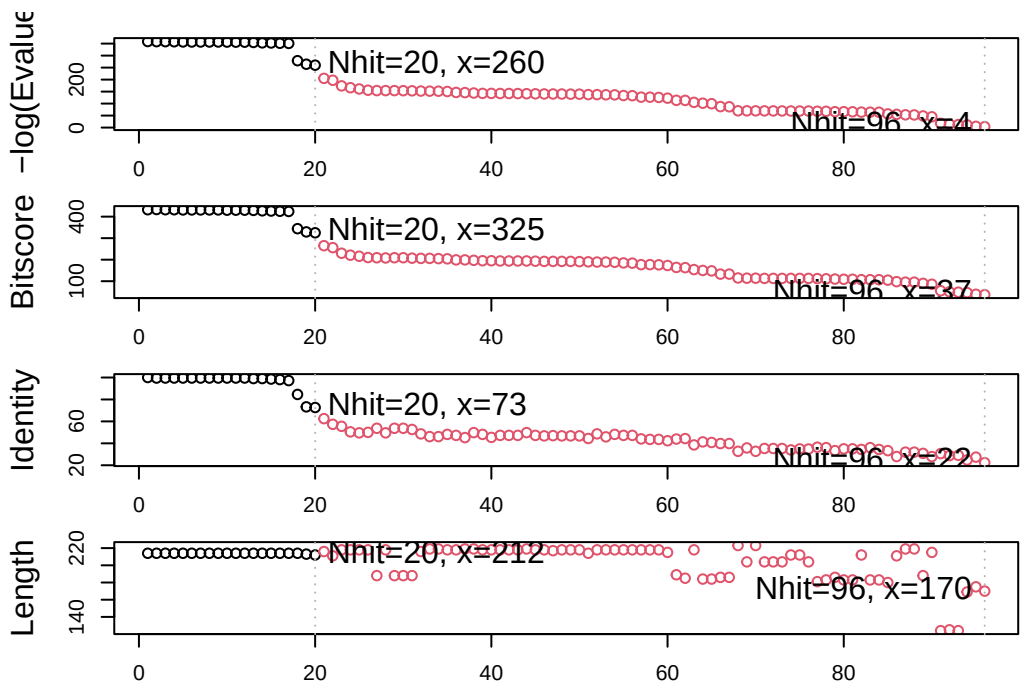
BLAST Search

```
b <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = Z7WG97Y6014
 ...
 Reporting 96 hits
```

```
hits <- plot.blast(b)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```




```
head(hits)
```

```
$hits
   pdb.id   acc      group
```

```
1  "1AKE_A" "1AKE_A" "1"
2  "8BQF_A" "8BQF_A" "1"
3  "4X8M_A" "4X8M_A" "1"
4  "6S36_A" "6S36_A" "1"
5  "9R6U_A" "9R6U_A" "1"
6  "9R71_A" "9R71_A" "1"
7  "8Q2B_A" "8Q2B_A" "1"
8  "8RJ9_A" "8RJ9_A" "1"
9  "6RZE_A" "6RZE_A" "1"
10 "4X8H_A" "4X8H_A" "1"
11 "3HPR_A" "3HPR_A" "1"
12 "1E4V_A" "1E4V_A" "1"
13 "5EJE_A" "5EJE_A" "1"
14 "1E4Y_A" "1E4Y_A" "1"
15 "3X2S_A" "3X2S_A" "1"
16 "6HAP_A" "6HAP_A" "1"
17 "6HAM_A" "6HAM_A" "1"
18 "8PVW_A" "8PVW_A" "1"
19 "4K46_A" "4K46_A" "1"
20 "4NP6_A" "4NP6_A" "1"

$pdb.id
 [1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
 [9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"

$acc
 [1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
 [9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"

$inds
 [1]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
[13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE FALSE FALSE
[25] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[37] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[49] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[61] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[73] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[85] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

Downloading the files:

```r
files <- get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8BQF.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8M.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download


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```

Aligning and Superposing structures:

Using pdb align - pdbaln()

```
pdbs <- pdbaln(files, fit = T, exefile = "msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/8BQF_A.pdb
pdbs/split_chain/4X8M_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/9R6U_A.pdb
pdbs/split_chain/9R71_A.pdb
```

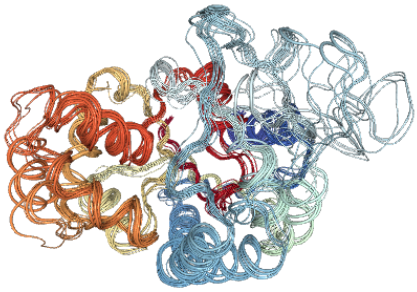
```
pdbs/split_chain/8Q2B_A.pdb
pdbs/split_chain/8RJ9_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/4X8H_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
```

```
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10   name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18   name: pdbs/split_chain/8PVW_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20   name: pdbs/split_chain/4NP6_A.pdb
```

Optional: Viewing our superposed structures

> Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```r
view.pdbs(pdbs, colorScheme = "residueIndex")
```



Annotate collected PDB structures

Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Vibrio cholerae O1 biovar El Tor str. N16961"
```

```
head(anno)
```

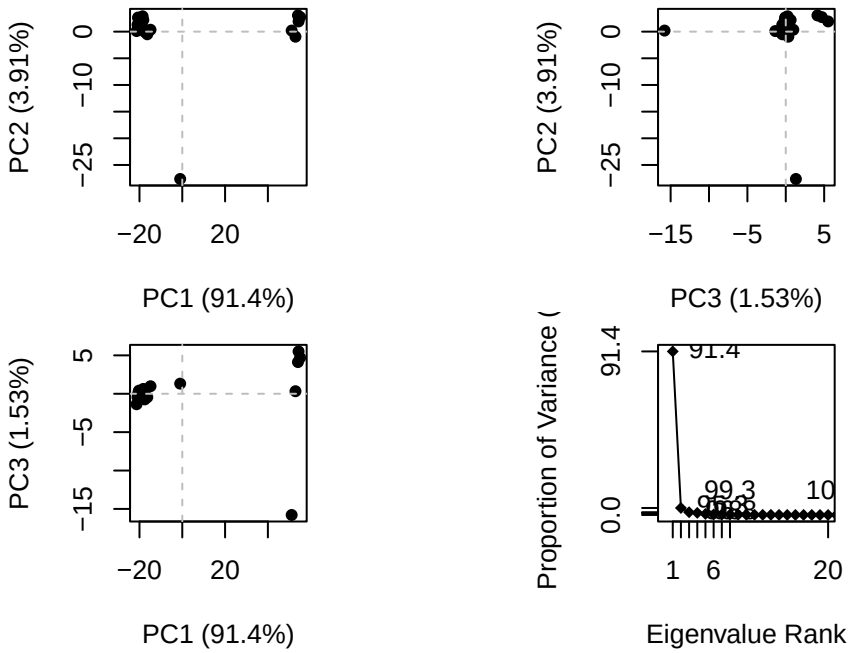
```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                 X-
ray
8BQF_A        8BQF       A           Protein         234                 X-
ray
4X8M_A        4X8M       A           Protein         214                 X-
ray
6S36_A        6S36       A           Protein         214                 X-
ray
9R6U_A        9R6U       A           Protein         214                 X-
ray
9R71_A        9R71       A           Protein         214                 X-
ray
       resolution       scopDomain                                        pfam
1AKE_A       2.00 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
8BQF_A       2.05             <NA> Adenylate kinase, active site lid (ADK_lid)
4X8M_A       2.60             <NA> Adenylate kinase, active site lid (ADK_lid)
6S36_A       1.60             <NA> Adenylate kinase, active site lid (ADK_lid)
9R6U_A       1.77             <NA> Adenylate kinase, active site lid (ADK_lid)
9R71_A       1.61             <NA> Adenylate kinase, active site lid (ADK_lid)
               ligandId                                          ligandName
1AKE_A              AP5                     BIS(ADENOSINE)-5'-PENTAPHOSPHATE
```

```
8BQF_A              AP5                     BIS(ADENOSINE)-5'-PENTAPHOSPHATE
4X8M_A             <NA>                                                 <NA>
6S36_A NA,MG (2),CL (3)        SODIUM ION,MAGNESIUM ION (2),CHLORIDE ION (3)
9R6U_A       AP5,NA,GOL BIS(ADENOSINE)-5'-PENTAPHOSPHATE,SODIUM ION,GLYCEROL
9R71_A              AP5                     BIS(ADENOSINE)-5'-PENTAPHOSPHATE
                 source
1AKE_A Escherichia coli
8BQF_A Escherichia coli
4X8M_A Escherichia coli
6S36_A Escherichia coli
9R6U_A Escherichia coli
9R71_A Escherichia coli

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
8BQF_A
4X8M_A
6S36_A
9R6U_A                                                                                    C
9R71_A                                                                                 Crys
                                                 citation rObserved   rFree
1AKE_A            Muller, C.W., et al. J Mol Biology (1992)   0.19600      NA
8BQF_A Scheerer, D., et al. Proc Natl Acad Sci U S A (2023)   0.22073 0.25789
4X8M_A              Kovermann, M., et al. Nat Commun (2015)   0.24910 0.30890
6S36_A                Rogne, P., et al. Biochemistry (2019)   0.16320 0.23560
9R6U_A             Mattsson, J., et al. Biochemistry (2025)        NA 0.22790
9R71_A             Mattsson, J., et al. Biochemistry (2025)   0.19600 0.24400
         rWork spaceGroup
1AKE_A 0.19600  P 21 2 21
8BQF_A 0.21882  P 2 21 21
4X8M_A 0.24630    C 1 2 1
6S36_A 0.15940    C 1 2 1
9R6U_A 0.19190  P 21 2 21
9R71_A 0.19300  P 21 21 2
```

PCA

> Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
pc.xray <- pca(pdbs)
plot(pc.xray)
```



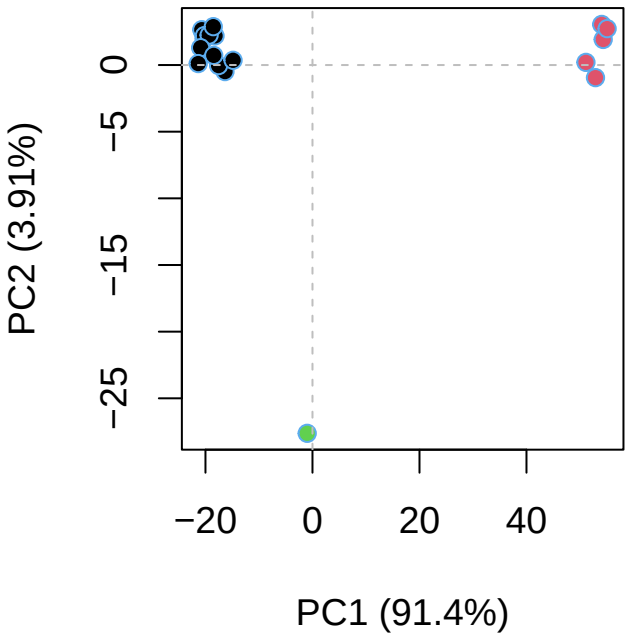
Calculating rmsd:

> Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 182 non NA positions
```

```
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k = 3)

plot(pc.xray, 1:2,col = "steelblue2", bg = grps.rd, pch = 21, cex = 1)
```



PCA Visualization

Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
pc1 <- mktrj(pc.xray, pc = 1, file = "pc1.pdb")
```

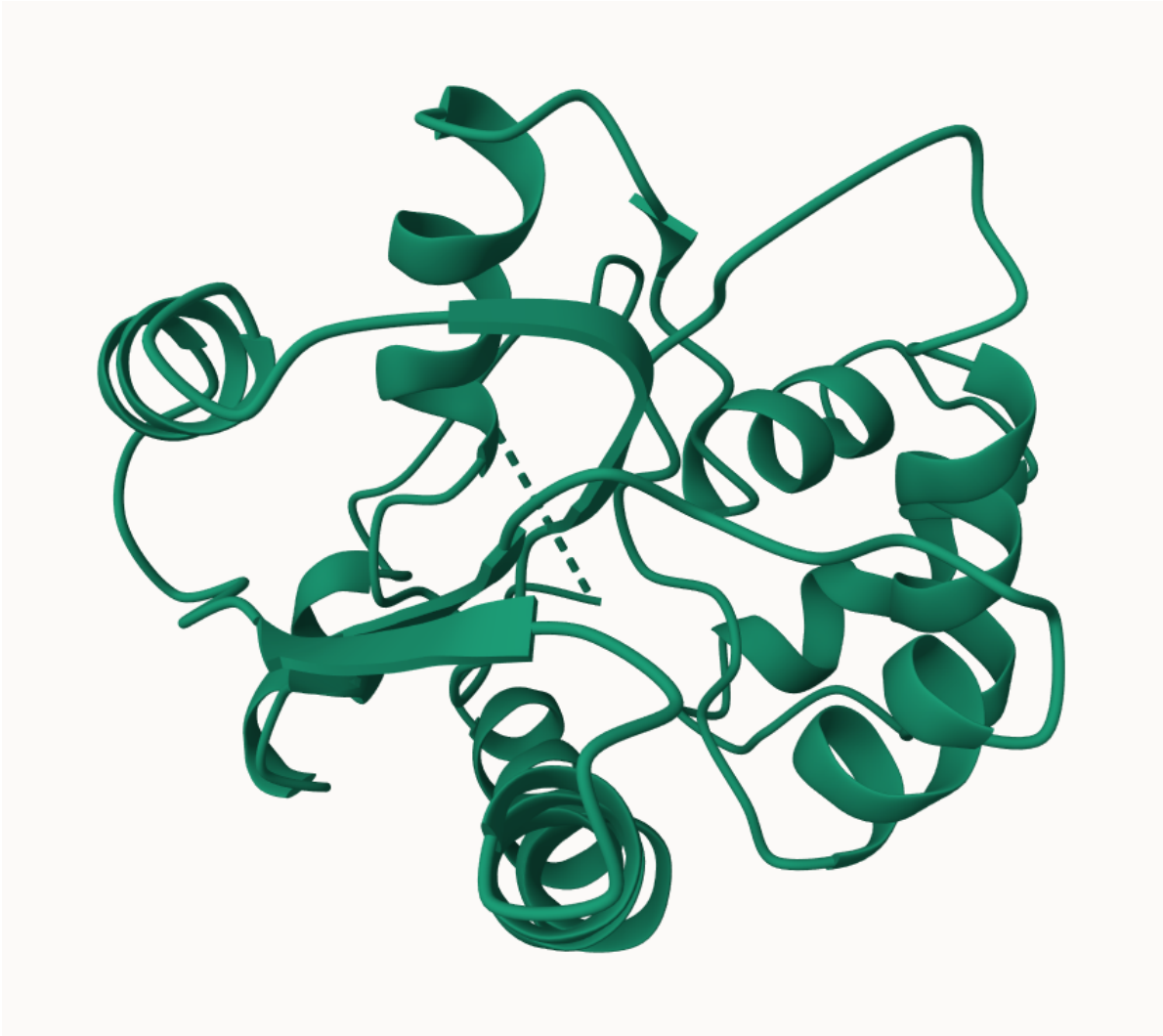


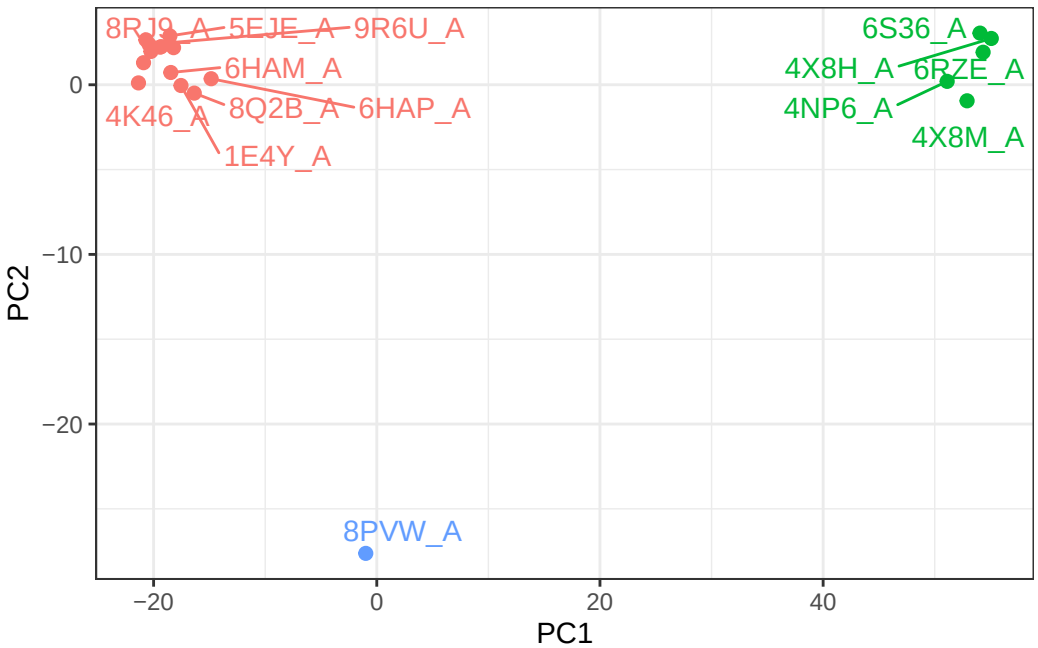
Figure 5: Output of visualizing PC1 in Mol* viewer - the animation showed 34 framed and moved through conformations of the molecule

Plotting with ggplot

> Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1 = pc.xray$z[ , 1],
```

```
                  PC2 = pc.xray$z[ , 2],
                  color = as.factor(grps.rd),
                  ids = ids)

p <- ggplot(df) +
  aes(PC1, PC2, col = color, label = ids) +
  geom_point(size = 2) +
  geom_text_repel(max.overlaps = 20) +
  theme_bw() +
  theme(legend.position = "none")
p
```



6. Normal mode analysis [optional]

NMA of all structures: > Note that this was created with 20 IDs found via a blast search, rather than the 13 that were pre-provided.

```
modes <- nma(pdbs)
```

```
Warning in nma.pdbs(pdbs): 8BQF_A.pdb might have missing residue(s) in structure:
   Fluctuations at neighboring positions may be affected.
```

```
Details of Scheduled Calculation:
  ... 20 input structures
  ... storing 540 eigenvectors for each structure
  ... dimension of x$U.subspace: ( 546x540x20 )
  ... coordinate superposition prior to NM calculation
  ... aligned eigenvectors (gap containing positions removed)
  ... estimated memory usage of final 'eNMA' object: 45.1 Mb


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```
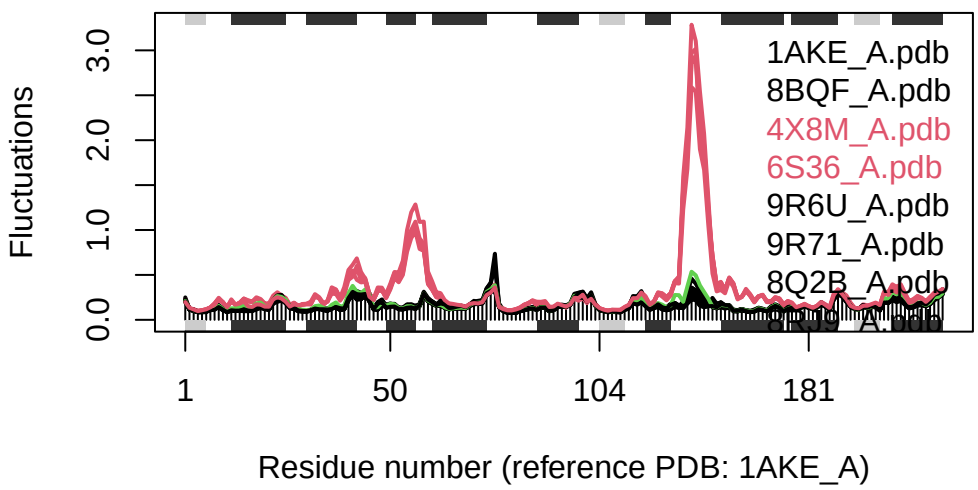

```
plot(modes, pdbs, col = grps.rd)
```

```
Extracting SSE from pdbs$sse attribute
```



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The black and colored lines occur at similar amino acid positions but differ greatly in relative fluctuations. They differ most around amino acid position ~55, and ~150. This probably represents two distinct binding domains that are flexible to allow for induced fit of the target ligand.